



Integrated assessment of air pollution dynamics and health impacts in Myanmar: Evidence from ground monitoring, satellite observations, and modeling

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ABSTRACT

Least Developed Countries (LDCs), such as Myanmar, experience disproportionately high air pollution burdens due to limited regulatory capacity, inadequate monitoring, and political instability. This study provides a multi-source assessment of air pollution dynamics and potential health impacts in Myanmar, combining ground-based monitoring, satellite observations, and statistical modeling to inform evidence-based policymaking. Ground-based data (2019–2024) from two monitoring stations in Yangon revealed an average PM_{2.5} concentration of 24.4 µg/m³ (SD = 18.6), with the highest seasonal mean in winter (39.4 µg/m³; November–February). Analysis of particulate matter (PM) components showed high contributions of sulfate (SO₄²⁻) in both winter and summer, followed by nitrate (NO₃⁻) in winter and Ca²⁺ in summer. A health impact assessment estimated that compliance with the WHO PM_{2.5} guideline (5 µg/m³) could have prevented 2,106 (95% confidence interval 345–3,818) premature deaths (aged ≥ 30) annually in Yangon during 2020–2021. A generalized additive model revealed that relative humidity and wind speed were significantly associated with PM_{2.5}, whereas the daily maximum 8 h average (MDA8) O₃ showed a limited meteorological association with relative humidity. National daily means of TROPOMI HCHO and NO₂ were strongly correlated (Pearson's $r = 0.754$; $p = 0.001$), and HCHO/NO₂ ratios indicated predominantly NO_x-sensitive regimes across Myanmar. These findings suggest that O₃ air quality can be substantially improved by reducing NO_x emissions. Despite Myanmar's limited data, this study provides a comprehensive spatiotemporal assessment of air pollution hotspots and their associated health burdens, thereby informing targeted emission control strategies.

1. Introduction

Myanmar, the largest country in mainland Southeast Asia, faces considerable air pollution challenges, particularly in its most urbanized and industrialized region, Yangon. According to the State of Global Air (2024) report (based on GBD, 2021 estimates), Myanmar was estimated to have approximately 101,600 premature deaths attributable to air pollution (Health Effects Institute, 2024). In Myanmar, major emission sources include biomass burning, household solid-fuel combustion, and mineral dust, while densely populated urban areas such as Yangon also experience relatively high vehicular and other combustion-related emissions (Zhang et al., 2022; Fodor and Ling, 2019). The country's tropical to subtropical monsoon climate is characterized by three

distinct seasons: hot summer (March to May), rainy monsoon (June to October), and cool winter (November to February), with significant regional variations in temperature and precipitation influenced by topography and the Southwest Monsoon (Red Cross Red Crescent Climate Centre, 2024). As the nation's most densely populated city, Yangon is situated on predominantly flat terrain and is bordered by multiple water bodies, with surrounding landscapes comprising rice paddies, brushland, and plantations.

In the tropical monsoon regions, fine particulate matter (PM_{2.5}) is strongly influenced by seasonal primary emissions (e.g., biomass burning) and secondary formation in combination with dry-season accumulation and monsoon wet scavenging (Khan et al., 2016; Lee et al., 2017; Kim Oanh et al., 2018). Specifically in humid conditions,

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the formation of sulfate and secondary organic aerosols is enhanced by aqueous-phase chemistry, leading to elevated PM_{2.5} air pollution (Gao et al., 2024; Huang et al., 2023; Chen et al., 2026). Ozone (O₃) air pollution is formed by the precursor gases of nitrogen oxides (NO_x) and volatile organic compounds (VOCs) through photochemical reactions. To understand spatially resolved precursor gas patterns and further O₃ sensitivity, the satellite observations of nitrogen dioxide (NO₂; a proxy of NO_x) and formaldehyde (HCHO; a proxy of VOCs) have been widely used to infer O₃ chemical regimes, either NO_x-limited or VOC-limited conditions (Duncan et al., 2010; Jin et al., 2020; Lee et al., 2023). For effective O₃ management, understanding regional O₃ chemical regimes is important because NO_x-limited and VOC-limited regimes require different control strategies and may vary across space and season (Souri et al., 2020; Wu et al., 2022). Health burdens related to air pollution exposures have been quantified by health impact assessment (HIA) frameworks (Bu et al., 2021; Apte et al., 2018). In data-scarce settings, due to limited individual- and population-level health data and thus epidemiological evidence, the assessment of air pollution exposures and health effects is often performed by the HIA, employing available data on air pollution concentrations, populations, baseline mortality rates, and exposure-response functions often derived from comprehensive epidemiological studies conducted in other countries, including the U.S. (Nguyen et al., 2022; Fold et al., 2020)

Existing studies from Yangon have demonstrated clear relationships between meteorological factors and air pollution levels, along with identifiable source signatures. A citywide analysis (2019–2021) showed that increases in rainfall, relative humidity, and dew point temperature were associated with decreases in PM_{2.5} and PM₁₀ concentrations, whereas higher temperatures tended to increase PM levels (Pyae and Kallawicha, 2024). At the Department of Meteorology and Hydrology (2018–2021), PM_{2.5} concentrations decreased with rainfall and temperature during the non-monsoon season and with relative humidity during the monsoon season, while wind speed had little influence (Sein et al., 2021). Health impact assessment frameworks provide a useful approach for translating air pollution exposure into avoidable mortality estimates (Li et al., 2016a; Liang et al., 2018). Early health impact assessments for Yangon using BenMAP-CE (2019) found that U.S.-based concentration–response coefficients overstated premature mortality, whereas Bangkok-based coefficients yielded more regionally appropriate estimates (Fold et al., 2020; Nandar et al., 2020).

Despite these previous efforts, air pollution studies in Myanmar remain fragmented. Many studies rely on single-site or short-term measurements, rarely integrate satellite-derived precursor data with ground-based observations, and seldom apply population-weighted exposure assessments using the updated WHO PM_{2.5} guideline (5 µg/m³) as a counterfactual benchmark. Therefore, this study provides a multi-source integrated assessment under data-limited conditions. Specifically, it aims to: (1) examine the associations between meteorological variables and PM_{2.5} and MDA8 O₃ using generalized additive models (GAMs); (2) assess Myanmar-wide O₃ precursor patterns using satellite NO₂ and HCHO observations and infer likely O₃ chemical regimes using HCHO/NO₂ ratios; and (3) estimate avoidable premature deaths in Yangon in relation to the WHO PM_{2.5} guideline. Collectively, these analyses provide the satellite-informed and health-oriented assessment of air pollution in Myanmar and support evidence-based discussion of monitoring expansion, pollution control, and public-health protection under limited data availability.

2. Methods

2.1. Ground measurements

Ground-based air quality data for Yangon were obtained from two fixed monitoring stations. Hourly PM_{2.5} concentrations were measured at the U.S. Embassy station (hereafter Station 1), which has been operational since April 2019 (data accessed via AirNow), and at the

Department of Meteorology and Hydrology station (hereafter Station 2), with data accessed through the Acid Deposition Monitoring Network in East Asia (EANET). (AirNow-Tech, 2025; EANET, 2025) These two sites are located approximately 5.9 km apart. In addition to PM_{2.5}, Station 1 provided surface O₃ concentrations, while Station 2 provided PM component measurements.

Station 1 is located on University Avenue Road near Inya Lake in an urban area of Yangon (16°49′31″N and 96°08′41″E). The PM_{2.5} measurements at this site were reported to be obtained using a BAM-1020 monitor based on beta attenuation (Pyae, 2021). The U.S. Embassy air-quality monitoring program states that the monitors follow common guidelines for siting, installation, operation, maintenance, and data quality assurance, including manufacturer specifications and U.S. EPA Ambient Air Quality Surveillance requirements (Embassy, 2019). Station 2 is located on Kaba-Aye Pagoda Road in Mayangon Township, Yangon. The EANET Data Report classifies it as an urban site (MMA001), located at 16°51′53″N and 96°09′13″E at an altitude of 21 m (Network Center for EANET, 2025). According to the EANET Technical Manual, PM_{2.5} monitoring using the β-ray absorption method is typically conducted with inlet heights of 3–10 m above ground (Network Center for EANET, 2013). However, the actual inlet height at the Yangon station was not reported in the publicly available metadata.

PM_{2.5} data were collected for 2019–2024 at Station 1 and for 2019–2023 at Station 2. Daily mean PM_{2.5} concentrations (µg/m³) were calculated only for days with at least 18 valid hourly measurements (i.e., ≥ 75% completeness). Daily mean PM_{2.5} concentrations were available for 89% and 87% of days during the study period at Stations 1 and 2, respectively. Daily MDA8 O₃ concentrations (ppb) were calculated for days meeting the validity criteria, yielding 514 valid days over 2019–2021. However, the availability of daily MDA8 O₃ data varied by year (2019: 180 days; 2020: 275 days; 2021: 59 days), which prevented the dataset from representing O₃ levels across all seasons in each year. Therefore, all valid data from 2019 to 2021 were retained for subsequent analyses to maximize sample size.

PM component concentrations were measured biweekly at Station 2 from 2012 to 2023 using the EANET filter pack method (Network Center for EANET, 2025). The dataset comprised bulk aerosol (non-size-selective) samples analyzed for ion-specified chemical composition, including sulfate (SO₄^{2−}), nitrate (NO₃[−]), ammonium (NH₄⁺), Ca²⁺, K⁺, Na⁺, Cl[−], and Mg²⁺. Annual data completeness was evaluated prior to further analysis. Data for 2022 were excluded due to less than 60% annual completeness. It should be noted that the PM component data at Station 2 were not size-selective for PM_{2.5} or PM₁₀. Additionally, other major species, such as organic carbon (OC) and elemental carbon (EC), were not measured at the site, restricting the analysis to comparisons among the available components only.

2.2. Meteorology

Meteorological variables were obtained from the ERA5-Land dataset, which provides hourly data at a 0.1° × 0.1° spatial resolution. (Copernicus Climate Change Service (C3S) 2025) These hourly data were aggregated to derive daily average meteorological parameters. The variables included 2 m dew point temperature (d2m; unit: K), 2 m air temperature (t2m; unit: K), surface pressure (sp; unit: Pa), total precipitation (tp; unit: m), and the eastward (u10; unit: m/s) and northward (v10; unit: m/s) wind components at 10 m above the surface. Wind speed (WS; unit: m/s) and relative humidity (RH; unit: %) were subsequently calculated, with RH estimated using the August–Roche–Magnus approximation as follows:

$$WS = \sqrt{u_{10}^2 + v_{10}^2}$$

$$RH = 100 \times \frac{\exp\left(\frac{17.625 \times d2m}{243.04 + d2m}\right)}{\exp\left(\frac{17.625 \times t2m}{243.04 + t2m}\right)}$$

The geographical locations of the two air quality monitoring stations, along with the spatial coverage of the meteorological data, are shown in Fig. 1.

2.3. Satellite observations

Gaseous air pollutant data were used to characterize the spatial patterns of air pollution across Myanmar from 2019 to 2024. Tropospheric vertical column densities of NO₂ and HCHO were obtained from Level-2 products of the Sentinel-5 Precursor (Sentinel-5P) TROPospheric Monitoring Instrument (TROPOMI) (unit: mol/m²). (NASA, 2025; NASA, 2025) To ensure data reliability and adequate spatial coverage, only retrievals with quality assurance (QA) values ≥ 0.5 were included. The spatial resolution of these products is 3.5 km $\times$ 5.5 km at nadir. Sentinel-5P, launched in October 2017, operates in a sun-synchronous orbit with an overpass time of approximately 13:30 local solar time at the equator (Veeffkind et al., 2012; Griffin et al., 2019; Theys et al., 2017; van Geffen et al., 2020).

After quality assurance filtering, valid TROPOMI NO₂ and HCHO observations were processed under the WGS84 geographic coordinate system (EPSG:4326), clipped to the Myanmar administrative boundary, and aggregated to a fixed 0.1° $\times$ 0.1° grid. Grid-cell mean values were calculated from available valid observations within each cell. No spatial interpolation or gap-filling was applied. For each overall or seasonal mean map, NO₂ and HCHO were first averaged separately within each 0.1° grid cell. The HCHO/NO₂ ratio was then calculated as the grid-cell mean HCHO divided by the grid-cell mean NO₂. Thus, the ratio was derived from aggregated grid-cell mean values rather than by averaging pixel-level HCHO/NO₂ ratios. Cell-wise differences between the 2019–2021 and 2022–2024 period means were also calculated to evaluate changes in NO₂, HCHO, and the HCHO/NO₂ ratio.

NO₂ and HCHO serve as proxies for NO_x and VOCs, respectively, which are major precursors of O₃ (Duncan et al., 2010; Lee et al., 2023).

Therefore, the HCHO/NO₂ ratio was used as an indicator of likely O₃ chemical regimes. Because satellite columns represent vertically integrated atmospheric information at the Sentinel-5P overpass time rather than continuous surface concentrations, the HCHO/NO₂ ratio was interpreted as a first-order spatial indicator of O₃ precursor conditions across Myanmar, providing an initial national-scale assessment under data-limited conditions.

2.4. Relationships between air pollutant concentrations and meteorological conditions

We examined the associations between meteorological variables and daily PM_{2.5} and MDA8 O₃ concentrations using GAMs, which allow for nonlinear effects of meteorological variables on pollutant concentrations. GAMs, a flexible regression framework that extends the generalized linear model, can capture complex nonlinear relationships between predictors and response variables (Hastie and Tibshirani, 1986). This flexibility makes GAMs particularly useful in air pollution studies, where pollutant concentrations often respond nonlinearly to meteorological and other temporal factors (Cheng et al., 2021).

The initial set of explanatory variables included 2 m air temperature, 2 m dew point temperature, surface pressure, total precipitation, wind speed, and relative humidity. To ensure robust model specification and mitigate multicollinearity, we performed Pearson correlation analysis for all pairs of explanatory variables. When two variables exhibited high correlation (Pearson $|r| > 0.5$), the variable more strongly correlated with the dependent variable was retained, while the other was excluded from the model. Following these diagnostics, relative humidity and wind speed were identified as statistically significant predictors ($p < 0.001$) and included in the final GAMs for PM_{2.5}. For MDA8 O₃, after collinearity screening and statistical testing within the GAM framework, only relative humidity was retained as the sole meteorological predictor, alongside year, month, and day of the week, in the final model. The objective of this analysis was to examine the observed associations between air pollution concentrations and key meteorological parameters under real-world atmospheric conditions, in which precursor gas emissions, photochemical reactions, or chemical regime characteristics were implicitly reflected. We did not aim to isolate independent causal effects

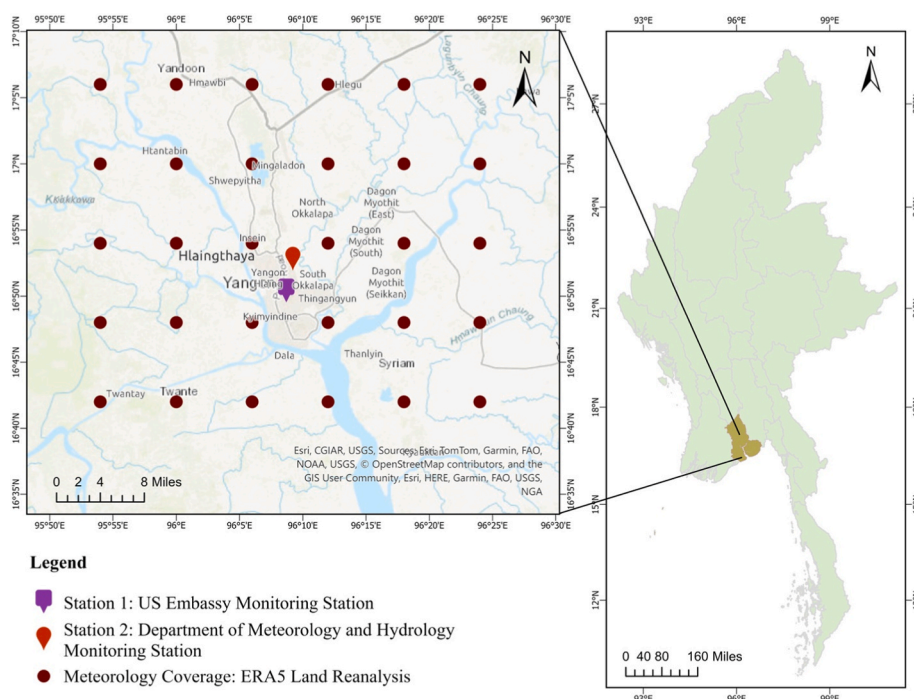


Fig. 1. Locations of ground-based air quality monitoring stations and meteorological data centroids (ERA5-Land) in the study area.

of the meteorological parameters on air pollution concentrations or to reconstruct air pollution concentrations, as addressed by air pollution estimation models, requiring a comprehensive set of predictors.

The general GAM formulation for PM_{2.5} was specified as:

$$g\{E(Y_t)\} = \beta_0 + \alpha \text{Year}_t + \gamma \text{Month}_t + \delta \text{DOW}_t + s(\text{WS}_t) + s(\text{RH}_t) + \varepsilon_t$$

where Y_t is the response variable (PM_{2.5}); $g(\cdot)$ denotes the link function; β_0 is the intercept; $s(\cdot)$ denotes smooth functions (splines) applied to the meteorological predictors; and α , γ , and δ are vectors of coefficients for the categorical variables year, month, and day of the week, respectively, capturing periodic cycles (month and day of the week) along with year-specific effects. For MDA8 O₃, the model followed the same general framework but retained only relative humidity among the meteorological variables.

Although daily total precipitation was excluded from the final PM_{2.5} GAM during the correlation-based variable screening, it was reintroduced in a sensitivity analysis because precipitation is a physically important wet-removal pathway for PM_{2.5} in tropical monsoon climates (Li et al., 2016b). Daily precipitation was included as a smooth term, $s(\text{tp})$, in an alternative PM_{2.5} GAM. For the MDA8 O₃ analysis, an exploratory precursor-proxy model was also tested by adding daily area-mean TROPOMI NO₂ and HCHO as smooth terms to examine whether satellite-based precursor proxies provided additional explanatory information to the daily station-level O₃ GAM. All GAMs were fitted using restricted maximum likelihood (REML), as implemented in the *mgcv* package in R (version 4.4.1) (Wood, 2025).

2.5. Health impact assessment of PM_{2.5}

The average annual number of premature deaths attributable to PM_{2.5} among individuals aged ≥ 30 years in Yangon during 2020–2021 was estimated using a standard log-linear function:

$$\Delta Y = Y_0 \cdot \text{Pop} \cdot (1 - e^{-\beta \Delta C})$$

where ΔY represents the estimated number of avoidable deaths if PM_{2.5} concentrations are reduced to the WHO guideline of 5 $\mu\text{g}/\text{m}^3$; Y_0 is the baseline mortality rate; Pop is the exposed population (aged ≥ 30 years); β is the exposure–response coefficient per unit change in PM_{2.5} concentration; and ΔC is the excess PM_{2.5} exposure above the WHO guideline, defined as the population-weighted mean PM_{2.5} concentration in Yangon for 2020–2021 minus 5 $\mu\text{g}/\text{m}^3$.

Population data were obtained from LandScan 2020–2021 raster datasets (30 arc-second resolution) developed by Oak Ridge National Laboratory (ORNL) and clipped to the Yangon administrative boundary using ArcGIS Pro 3.4.3 (Esri, Redlands, CA, USA) (Rose et al., 2020; Sims et al., 2022). Population counts in each grid cell were linked to PM_{2.5} concentrations measured at the nearest ground monitoring site (Station 1 or Station 2), while recognizing that this monitor-based approach only with two stations might not fully capture within-city spatial variability. Population counts were then aggregated by assigned station to obtain the total population per station. These totals were multiplied by the proportion of individuals aged ≥ 30 years, obtained from the Yangon Region Population Department, to estimate the population aged ≥ 30 years for each site and year (Myanmar Inter-censal Survey, 2019). For each year (2020 and 2021), annual mean PM_{2.5} concentrations at the two monitoring stations were paired with their corresponding populations to calculate the population-weighted average PM_{2.5} concentration. The two-year mean population-weighted concentration was then calculated as the average of the values for 2020 and 2021.

Baseline mortality data for Myanmar were obtained from the Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) database, as Yangon-specific mortality statistics were unavailable. (Global Burden of Disease Study, 2021) Age-specific non-accidental mortality rates (aged ≥ 30 years) were derived by subtracting injury-related (external-cause) deaths from all-cause deaths within each

age group and year, and then dividing the result by the corresponding age-specific population. These rates were expressed as annual deaths per 100,000 population and averaged over 2020–2021. The mean baseline mortality rate for 2020–2021 was 1,518 per 100,000 population, which was converted to 0.01518 and used as Y_0 in the health impact calculations.

The concentration–response coefficient (β) for all-cause non-accidental mortality attributable to ambient PM_{2.5} in Yangon was adopted from Fold et al. (2020), an epidemiological analysis of PM_{2.5}-attributable mortality in Bangkok, Thailand (Fold et al., 2020). The log-linear coefficient was $\beta = 0.001743$ per 1 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} (SE = 0.0007458). In the absence of Myanmar-specific coefficient estimates, this value was used as the most contextually appropriate regional proxy. To maintain comparability with Fold et al., whose analysis targeted individuals aged 30–99 years, our assessment focused on individuals aged ≥ 30 years, despite the IHME GBD data extending only to ages ≥ 95 years and the Yangon population statistics extending to ≥ 90 years.

For a sensitivity analysis, the avoidable mortality estimate was recalculated using an alternative concentration–response coefficient from Li et al. (2023), a recent nationwide cohort study from the ChinaHEART project (Li et al., 2023). The study enrolled adults aged 35–75 years from 31 provinces of China and reported an all-cause mortality hazard ratio of 1.02 (95% CI: 1.012–1.028) per 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5}. This corresponded to $\beta = 0.001980$ per 1 $\mu\text{g}/\text{m}^3$, with a 95% CI of 0.001193–0.002762 per 1 $\mu\text{g}/\text{m}^3$.

3. Results and discussion

3.1. Temporal patterns of ambient air pollution and meteorology

Table 1 summarizes daily PM_{2.5} concentrations at the two Yangon stations by year and month, along with MDA8 O₃ at Station 1 and meteorological conditions (temperature, relative humidity, and wind speed). At Stations 1 and 2, daily PM_{2.5} concentrations ranged from 0.0 to 88.7 $\mu\text{g}/\text{m}^3$ and from 0.77 to 106.4 $\mu\text{g}/\text{m}^3$, respectively. No clear annual trends in PM_{2.5} concentrations were observed at either station. Seasonal variations in PM_{2.5} were pronounced at both stations, with maximum concentrations in February (winter; 48.8–53.8 $\mu\text{g}/\text{m}^3$) and minimum concentrations in July (monsoon; 4.4–8.6 $\mu\text{g}/\text{m}^3$). Monthly PM_{2.5} concentrations from January to March were higher at Station 1, whereas the opposite pattern was observed during the remaining months. These differences may reflect local environmental and emission conditions around Station 2, although the differences in monitoring networks need to be also considered. MDA8 O₃ concentrations at Station 1 ranged from 1.0 to 70.0 ppb during 2019–2021. Annual means were comparable in 2019 and 2020 but higher in 2021. However, substantially lower data availability in 2021 ($n = 59$ days, compared to 180 days in 2019 and 275 days in 2020) limited robust interannual comparisons. A clear seasonal cycle was observed, with elevated MDA8 O₃ concentrations during the dry season and marked reductions during the monsoon period. Meteorological conditions were consistently warm and humid, with mean values of 27.2 °C for temperature and 74.5% for relative humidity. Monthly mean meteorological conditions ranged from 24.9 to 30.7 °C for temperature, 54.5% to 90.2% for relative humidity, and 1.2 to 2.0 m/s for wind speed.

Fig. 2 illustrates the diurnal and day-of-week patterns of PM_{2.5} and O₃ measured in Yangon. The diurnal cycle of PM_{2.5} showed peaks in the morning (06:00–09:00) and evening (19:00–22:00), with a distinct midday minimum. At Stations 1 and 2, the morning peaks were higher than the evening peaks, and both peaks were higher at Station 2 than the corresponding peaks at Station 1. The higher PM_{2.5} peaks and their broader range at Station 2 indicate greater surrounding emission intensity than at Station 1, possibly reflecting denser traffic activity (i.e., its location next to a major road corridor, a busy intersection, and a main bus route (0.09–0.12 km away)) (Sein et al., 2021). Average PM_{2.5} concentrations differed slightly between weekdays and weekends: at

Table 1

Temporal variations in daily PM_{2.5} concentrations ($\mu\text{g}/\text{m}^3$) measured at two Yangon ground stations (Station 1: 2019–2024; Station 2: 2019–2023), MDA8 O₃ (ppb) at Station 1 (2019–2021), and meteorological variables (temperature ($^{\circ}\text{C}$), relative humidity (%), and wind speed (m/s)) for 2019–2024. Means and standard deviations are shown.

Category	PM _{2.5} Station 1	PM _{2.5} Station 2	MDA8 O ₃ Station 1	Temp	RH	WS
Overall	22.3 (19.5)	25.4 (18.4)	14.0 (9.2)	27.2 (2.1)	74.5 (15.0)	1.6 (0.7)
Year						
2019	19.6 (16.3)	23.4 (15.2)	13.7 (8.7)	27.3 (2.2)	73.1 (15.5)	1.7 (0.7)
2020	25.7 (22.7)	21.9 (15.6)	13.2 (10.1)	27.2 (2.0)	72.6 (15.9)	1.6 (0.6)
2021	22.0 (20.4)	28.3 (24.9)	18.7 (3.3)	26.9 (1.8)	75.4 (14.1)	1.5 (0.7)
2022	20.5 (16.8)	26.3 (17.4)	–	27.0 (1.9)	76.0 (14.0)	1.5 (0.7)
2023	21.9 (19.4)	27.0 (16.9)	–	27.4 (2.1)	75.0 (14.4)	1.6 (0.7)
2024	23.5 (19.6)	–	–	27.4 (2.3)	74.9 (16.0)	1.7 (0.7)
Season						
Winter						
November	23.0 (8.6)	28.2 (11.0)	17.0 (6.5)	27.0 (0.9)	78.1 (6.9)	1.2 (0.6)
December	35.9 (11.3)	37.7 (12.1)	17.9 (5.0)	25.6 (1.4)	68.7 (6.9)	1.4 (0.6)
January	47.9 (12.6)	44.1 (15.4)	16.8 (1.9)	24.9 (1.2)	62.9 (7.6)	1.3 (0.6)
February	53.8 (13.7)	48.8 (19.6)	20.7 (3.2)	26.2 (1.4)	54.5 (7.4)	1.3 (0.5)
Summer						
March	44.9 (12.1)	42.6 (16.3)	16.6 (5.0)	28.9 (1.0)	54.8 (7.5)	1.6 (0.5)
April	34.2 (13.4)	34.5 (11.4)	18.4 (5.9)	30.7 (1.5)	59.7 (8.3)	1.8 (0.5)
May	16.5 (9.2)	17.8 (7.7)	12.1 (3.8)	29.9 (2.1)	72.1 (11.8)	2.0 (0.8)
Monsoon						
June	5.8 (3.8)	10.2 (3.9)	6.4 (12.9)	26.9 (0.9)	88.4 (4.6)	2.0 (0.6)
July	4.4 (2.9)	8.6 (3.5)	23.3 (17.3)	26.6 (0.8)	89.2 (3.9)	2.0 (0.6)
August	5.3 (3.3)	10.6 (8.1)	8.7 (7.1)	26.4 (0.8)	90.2 (3.6)	2.0 (0.7)
September	5.5 (5.3)	11.9 (7.5)	7.3 (7.4)	26.6 (0.9)	88.9 (4.1)	1.5 (0.6)
October	10.5 (8.5)	14.3 (6.0)	10.8 (6.0)	27.1 (0.9)	85.3 (4.8)	1.2 (0.8)

Station 1, 22.3 versus 22.2 $\mu\text{g}/\text{m}^3$; at Station 2, 25.7 versus 24.6 $\mu\text{g}/\text{m}^3$. For O₃, the diurnal cycle exhibited minima during the late night and early morning hours (00:00–06:00) and peak concentrations in the afternoon (13:00–16:00), reflecting photochemical production driven by solar radiation. Mean O₃ concentrations during weekends (13.7 ppb) were slightly lower than those on weekdays (14.1 ppb), although the difference (weekend-to-weekday O₃ ratio = 0.97) was not statistically significant.

Fig. 3 presents the seasonal mean concentrations and relative contributions of PM components, including SO_4^{2-} , NO_3^- , Cl^- , NH_4^+ , Na^+ , K^+ , Mg^{2+} , and Ca^{2+} , measured at Station 2. The sum of all PM component concentrations was 7.02 $\mu\text{g}/\text{m}^3$ in winter, followed by summer (6.01 $\mu\text{g}/\text{m}^3$), with the lowest concentrations observed during the monsoon season (3.39 $\mu\text{g}/\text{m}^3$). Because these samples represented bulk aerosol composition, rather than size-selective PM_{2.5} or PM₁₀ composition, the summed ionic concentrations were not used for direct mass-balance comparison with PM_{2.5}. Instead, the analysis focused on seasonal variation and relative contributions among the measured ionic components. Among the measured PM components, sulfate (SO_4^{2-}) was the dominant inorganic component in both winter and summer, contributing 31.0% (2.18 $\mu\text{g}/\text{m}^3$) and 35.7% (2.15 $\mu\text{g}/\text{m}^3$) of the total measured PM component concentrations, respectively. In winter, sulfate was followed by nitrate (NO_3^- ; 17.0%, 1.19 $\mu\text{g}/\text{m}^3$), calcium (Ca^{2+} ; 16.1%, 1.13 $\mu\text{g}/\text{m}^3$), and potassium (K^+ ; 14.8%, 1.04 $\mu\text{g}/\text{m}^3$), indicating notable influences from secondary inorganic aerosol formation, dust, and biomass burning. During summer, sulfate dominance persisted, with substantial contributions from Ca^{2+} (20.6%, 1.24 $\mu\text{g}/\text{m}^3$), ammonium (NH_4^+ ; 10.3%, 0.62 $\mu\text{g}/\text{m}^3$), and nitrate (NO_3^- ; 10.1%, 0.61 $\mu\text{g}/\text{m}^3$), reflecting the continued importance of secondary inorganic aerosols under dry-season conditions. During the monsoon season, total PM component concentrations were substantially reduced, reflecting efficient wet scavenging. In contrast to winter and summer, sodium (Na^+) became the largest single contributor during the monsoon season (21.9%, 0.74 $\mu\text{g}/\text{m}^3$), followed by sulfate (20.1%, 0.68 $\mu\text{g}/\text{m}^3$). Potassium (K^+ ; 12.7%, 0.43 $\mu\text{g}/\text{m}^3$) and chloride (Cl^- ; 11.3%, 0.38 $\mu\text{g}/\text{m}^3$) also increased in relative importance despite lower absolute concentrations, consistent with enhanced marine influence and the preferential removal of secondary inorganic components during periods of frequent precipitation.

These findings are broadly consistent with a previous year-long study of PM₁₀ in Yangon, which reported elevated secondary aerosols and dust-related species during winter and summer, and a pronounced reduction in these species during the monsoon (Sricharoenvech et al., 2020). Furthermore, elevated K^+ levels during the dry months are consistent with possible biomass burning influence, consistent with source apportionment markers previously identified in the Yangon urban environment (Sricharoenvech et al., 2020). Sulfate consistently accounted for approximately one-third of the total measured PM component concentrations during the dry seasons, which may reflect contributions from secondary sulfate formation and regional fuel-related sulfur emissions. This interpretation is consistent with the regulatory context, characterized by prolonged reliance on high-sulfur diesel fuels and limited emission-control technologies under pre-Euro vehicle standards (Jeong, 2022). Although imported Euro IV-equivalent diesel (50 ppm sulfur) is currently used to meet fuel demand, Myanmar has not yet implemented a domestic production mandate for low-sulfur diesel; thus, legacy fuel use and fleet composition likely continue to influence the observed sulfate pattern (Jeong, 2022). However, these source-related inferences from our non-size-selective PM component data remain qualitative because OC, EC, trace metals, size-resolved composition, and emission inventory data were unavailable. Therefore, the PM component results should be interpreted as evidence of seasonal variation in the measured ionic components, rather than a complete PM_{2.5} chemical mass balance or formal source attribution.

3.2. Relationships between air pollutant concentrations and meteorological conditions

The final GAMs yielded adjusted R² values of 0.824 at Station 1 and 0.675 at Station 2, indicating that the models explained a substantial proportion of daily PM_{2.5} variability. As shown in Fig. 4A, relative humidity was a significant nonlinear smooth term at Station 1 [$p < 0.001$, effective degrees of freedom (EDF) = 5.5]. The concavity diagnostics showed that the smooth function of relative humidity had relatively high concavity mainly with the categorical term of month, reflecting the strong seasonal structure of relative humidity. On the other hand,

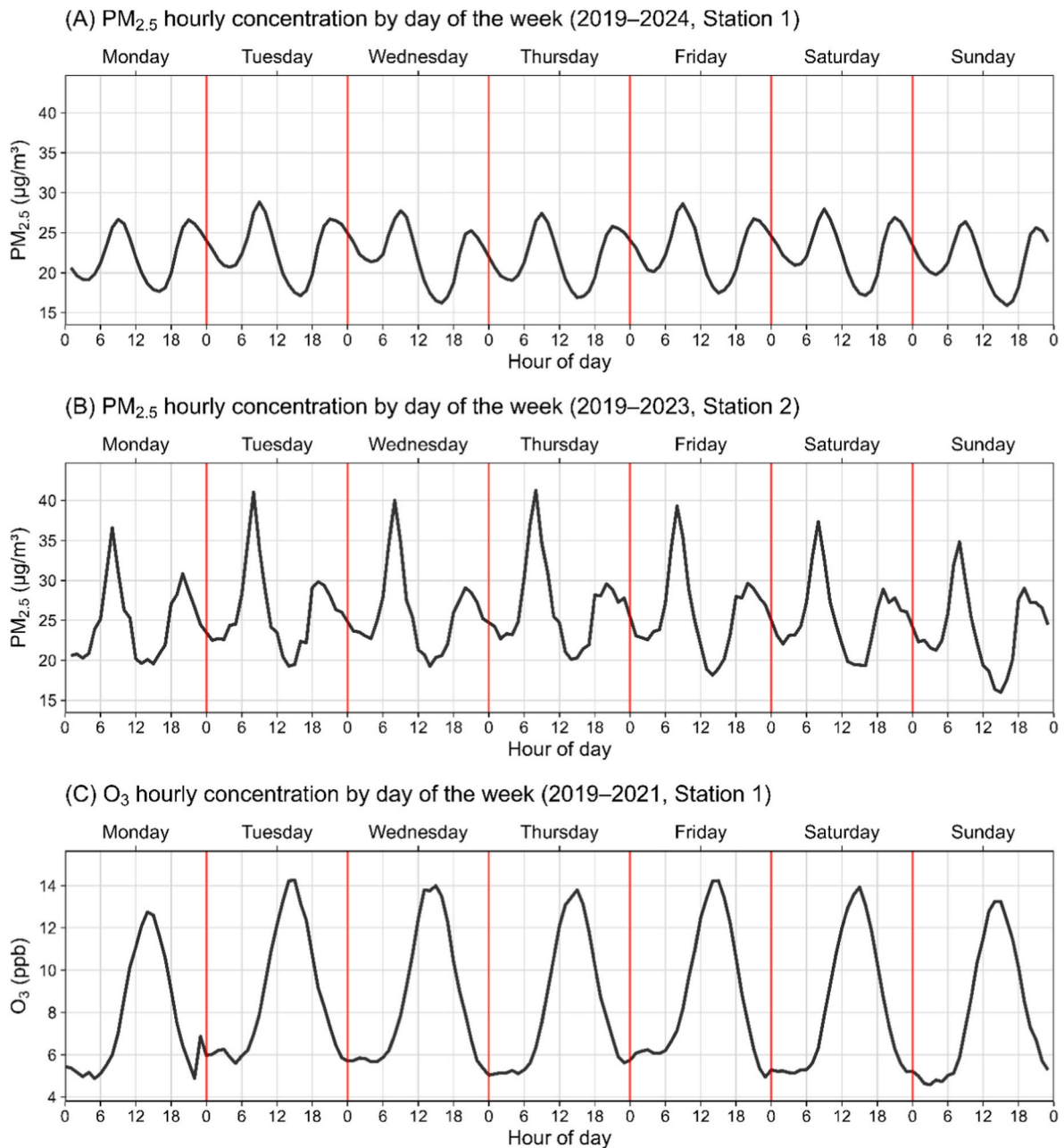


Fig. 2. Diurnal and day-of-week variations in O₃ and PM_{2.5} concentrations in Yangon: (A) hourly PM_{2.5} concentrations at Station 1 (2019–2024), (B) PM_{2.5} hourly concentrations at Station 2 (2019–2023), and (C) O₃ hourly concentrations (2019–2021). Vertical red lines indicate daily boundaries. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the smooth function of wind speed showed moderate and low concavity with the temporal terms and that of relative humidity at both stations, respectively. The partial effect on PM_{2.5} was highest at low humidity (~40%) and declined steadily as relative humidity increased, with a steeper decrease above approximately 55–60%, consistent with wet removal under more humid conditions. Wind speed was also statistically significant and showed a moderately nonlinear association ($p < 0.001$, EDF = 3.0), with the partial effect on PM_{2.5} decreasing sharply at low speeds (0–2 m/s) before stabilizing, highlighting the critical role of atmospheric dispersion.

As shown in Fig. 4B, the GAM results for Station 2 indicated a more complex nonlinear relationship with relative humidity ($p < 0.001$, EDF = 7.1). Unlike the monotonic decline observed at Station 1, the partial effect of relative humidity on PM_{2.5} reached a localized peak at approximately 55% of the relative humidity. Within the 50–70% of the

relative humidity range, the response exhibited noticeable fluctuations before abruptly decreasing beyond 70% of the relative humidity. This pattern may reflect the combined influence of local emissions, humidity-related particle processes, and seasonal meteorological conditions. Wind speed also showed a significant, moderately nonlinear effect ($p < 0.001$, EDF = 2.9), with the partial effect on PM_{2.5} decreasing as wind speed increased from 0 to about 3 m/s. Beyond this range, the curve flattened and then slightly increased at the highest wind speeds, suggesting a possible threshold where wind-driven resuspension begins to offset mechanical dispersion.

In a sensitivity analysis, we reintroduced the smooth function of precipitation into the GAMs to evaluate physically important wet-removal pathway for PM_{2.5}. However, adding precipitation into the model did not improve the model performance or alter the smooth functions of relative humidity and wind speed (Text S1). In addition,

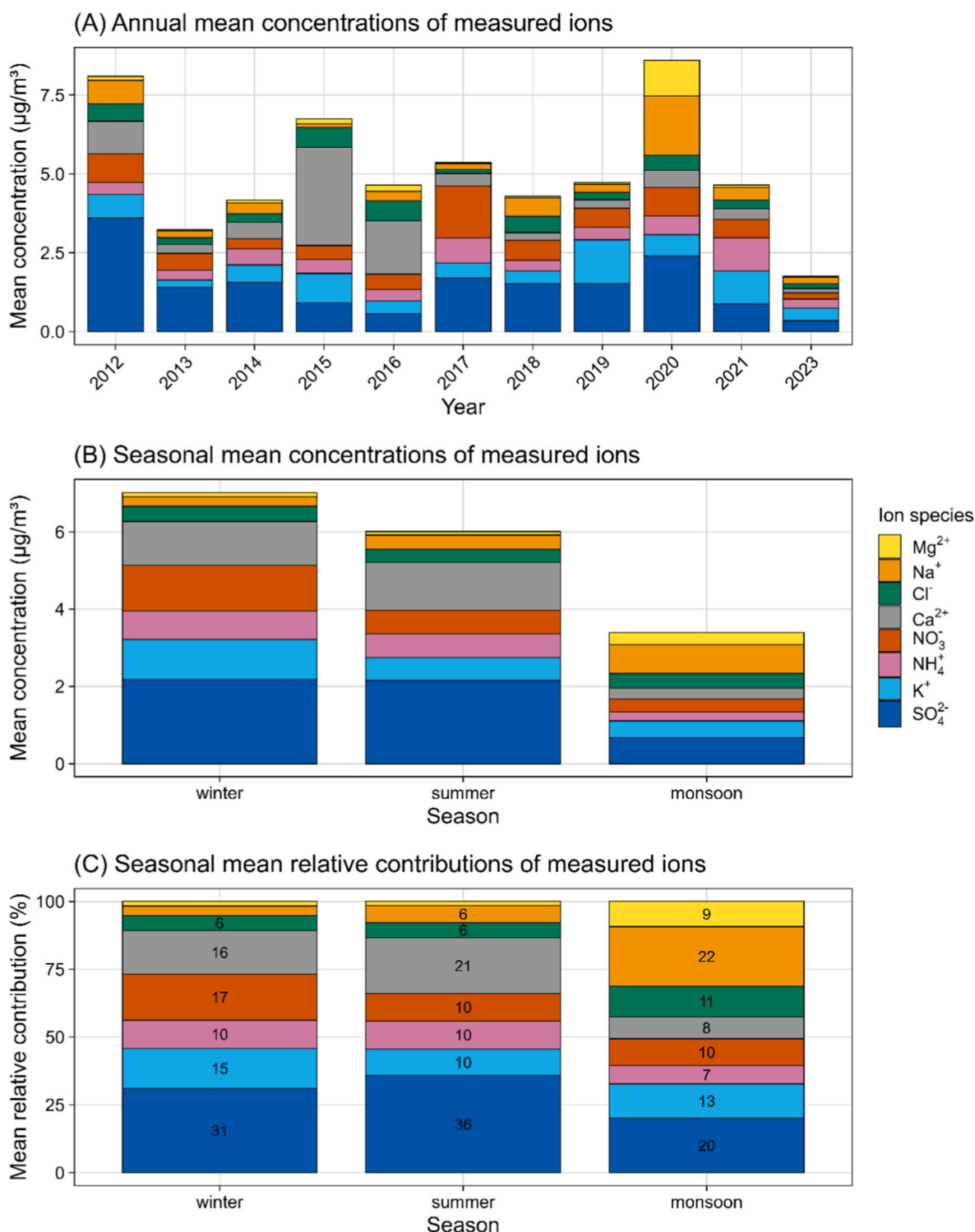


Fig. 3. Annual and seasonal mean concentrations and relative contributions of PM components measured at Station 2: (A) annual mean concentrations of individual PM components ($\mu\text{g}/\text{m}^3$), (B) seasonal mean concentrations of individual PM components ($\mu\text{g}/\text{m}^3$), and (C) seasonal mean relative contributions (%) of each PM component.

adding precipitation increased concurrency with relative humidity. Therefore, precipitation was not retained in the final GAMs for PM_{2.5} despite the fact that precipitation may play a more prominent role in explaining the PM_{2.5} variability during the monsoon season.

The final GAM for MDA8 O₃ concentrations yielded an adjusted R² of 0.339 at Station 1 ($n = 514$), with limited data availability in 2021 ($n = 59$). This limited explanatory power may be attributed to the greater

proportion of missing O₃ measurements and the complexity of atmospheric processes, which strongly depend on precursor gases (e.g., NO_x and VOCs) not included in the GAM. Relative humidity remained the only statistically significant meteorological smooth term ($p < 0.001$, EDF = 1.27). The EDF value close to 1 indicates that the relative humidity–O₃ association was nearly linear. While the smooth function of relative humidity showed relatively high concurrency mainly with the

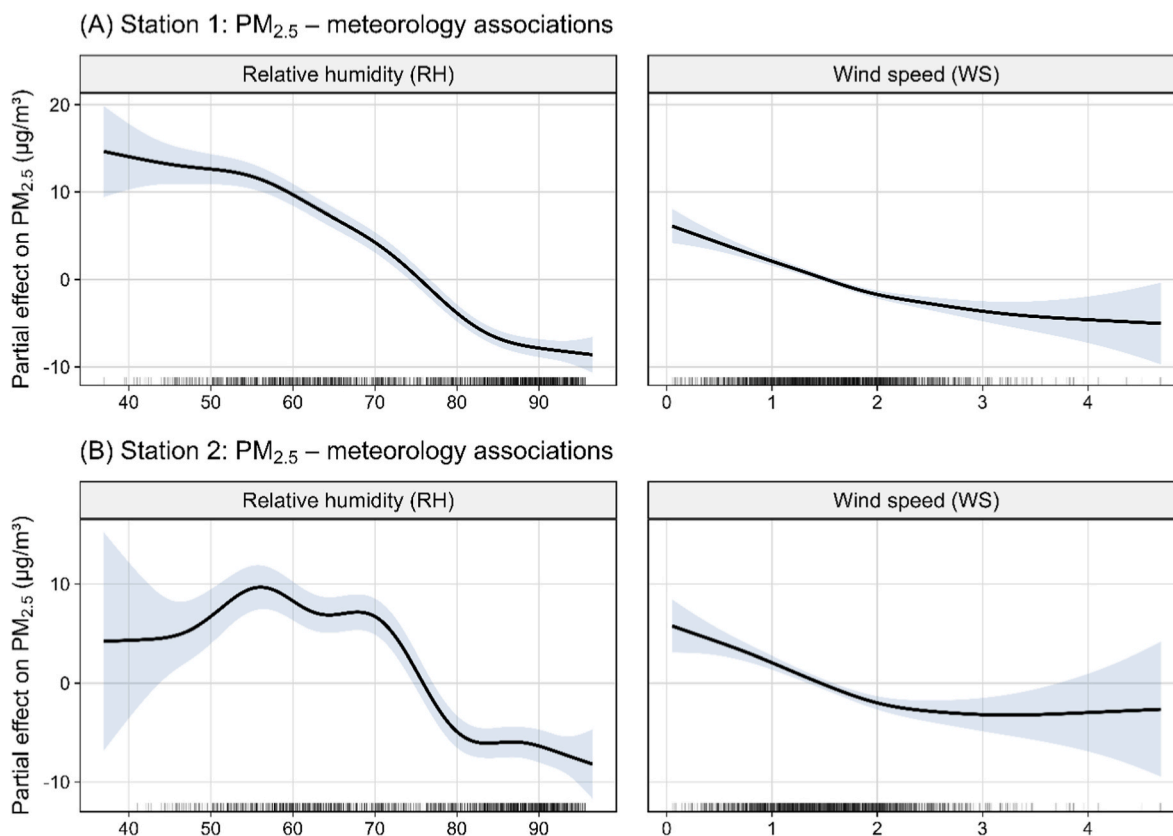


Fig. 4. Relationships between $\text{PM}_{2.5}$ concentrations and meteorological parameters at Stations 1 and 2, after accounting for nonlinearity and adjusting for year, month, and day of the week in the GAM. Black solid curves represent centered smooth effects ($\mu\text{g}/\text{m}^3$), with 95% confidence bands (blue shaded areas). Short vertical lines on the x-axis indicate individual observations. Relative humidity and wind speed are expressed in units of % and m/s, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

categorical term of month, MDA8 O_3 was generally higher at lower relative humidity and declined nearly linearly with increasing relative humidity. Despite the model residuals generally centered around zero, several intermittent high residuals were observed, which can be attributed to episodic photochemical reactions, precursor gas emissions, regional transport, and/or air ventilation that may not be comprehensively explained by routine meteorological parameters (Fig. S1). Hence, the meteorology- O_3 associations were not interpreted as independent causal meteorological effects on O_3 .

An exploratory precursor-proxy model for MDA8 O_3 was also constructed by adding daily area-mean TROPOMI NO_2 and HCHO values. Because this model required concurrent availability of MDA8 O_3 , relative humidity, and TROPOMI NO_2 and HCHO data, the sample size decreased from 514 to 394 days, potentially producing unstable estimates. Adding TROPOMI NO_2 and HCHO into the model slightly increased the adjusted R^2 from 0.313 to 0.336. However, compared with the final O_3 GAM using all valid O_3 observations ($n = 514$), the adjusted R^2 slightly decreased from 0.339 to 0.336. The smooth term for TROPOMI NO_2 was statistically significant ($p = 0.00375$), whereas TROPOMI HCHO was not significant ($p = 0.208$), and relative humidity remained statistically significant ($p = 0.000288$). The limited explanatory contribution of TROPOMI NO_2 and HCHO may reflect the fact that these variables represent vertically integrated satellite column observations rather than continuous surface precursor concentrations at the monitoring site.

3.3. Spatial distributions of satellite-based NO_2 , HCHO, and O_3 formation sensitivities

Fig. 5A illustrates the spatial patterns of TROPOMI tropospheric

column NO_2 over Myanmar from 2019 to 2024 on a 0.1° (~ 10 km) grid. Across all seasons, the national mean tropospheric column NO_2 was $14.9 \mu\text{mol}/\text{m}^2$ ($\text{SD} = 3.1$), with clear seasonal and spatial variability. Average NO_2 levels peaked in summer ($24.6 \mu\text{mol}/\text{m}^2$, $\text{SD} = 8.6$), likely due to active biomass burning, particularly in eastern Myanmar, and declined sharply during the monsoon ($11.6 \mu\text{mol}/\text{m}^2$, $\text{SD} = 2.1$) because of increased wet deposition and atmospheric cleansing (Oo et al., 2020). In each season, the highest NO_2 levels were observed in Yangon and the Mandalay Plain, which represent Myanmar's primary urban-industrial corridors. These elevated NO_2 levels corresponded to areas of dense population, intense transportation activity, and industrial development. Yangon, Myanmar's commercial capital, and Mandalay, the country's second-largest urban center and the economic hub of Upper Myanmar, both exhibit substantial urbanization, extensive industrial zones, and well-developed transport corridors that sustain NO_x emissions (Figs. S2 and S3). For instance, urban townships in Mandalay, such as Maha Aung Myay and Chanmyatharzi, have population densities exceeding 11,000–16,000 people per km^2 , with steady growth over recent decades (Department of Population, 2017). By contrast, persistently low NO_2 levels were observed in the northern mountainous and forested regions, the outer Ayeyarwady Delta, and the remote coastal and island areas of Tanintharyi, where anthropogenic sources are limited, and cleaner maritime air masses prevail. These spatial variations in NO_2 levels were strongly influenced by the distribution of emission sources, with cropland and built-up areas generally exhibiting higher NO_2 concentrations, whereas forested areas corresponded to lower levels.

Fig. 5B displays the spatial distribution of TROPOMI tropospheric column HCHO over Myanmar from 2019 to 2024 on a 0.1° (~ 10 km) grid. Across all seasons, the average HCHO level was $159.4 \mu\text{mol}/\text{m}^2$ ($\text{SD} = 26.7$), with pronounced seasonal variation. The highest HCHO

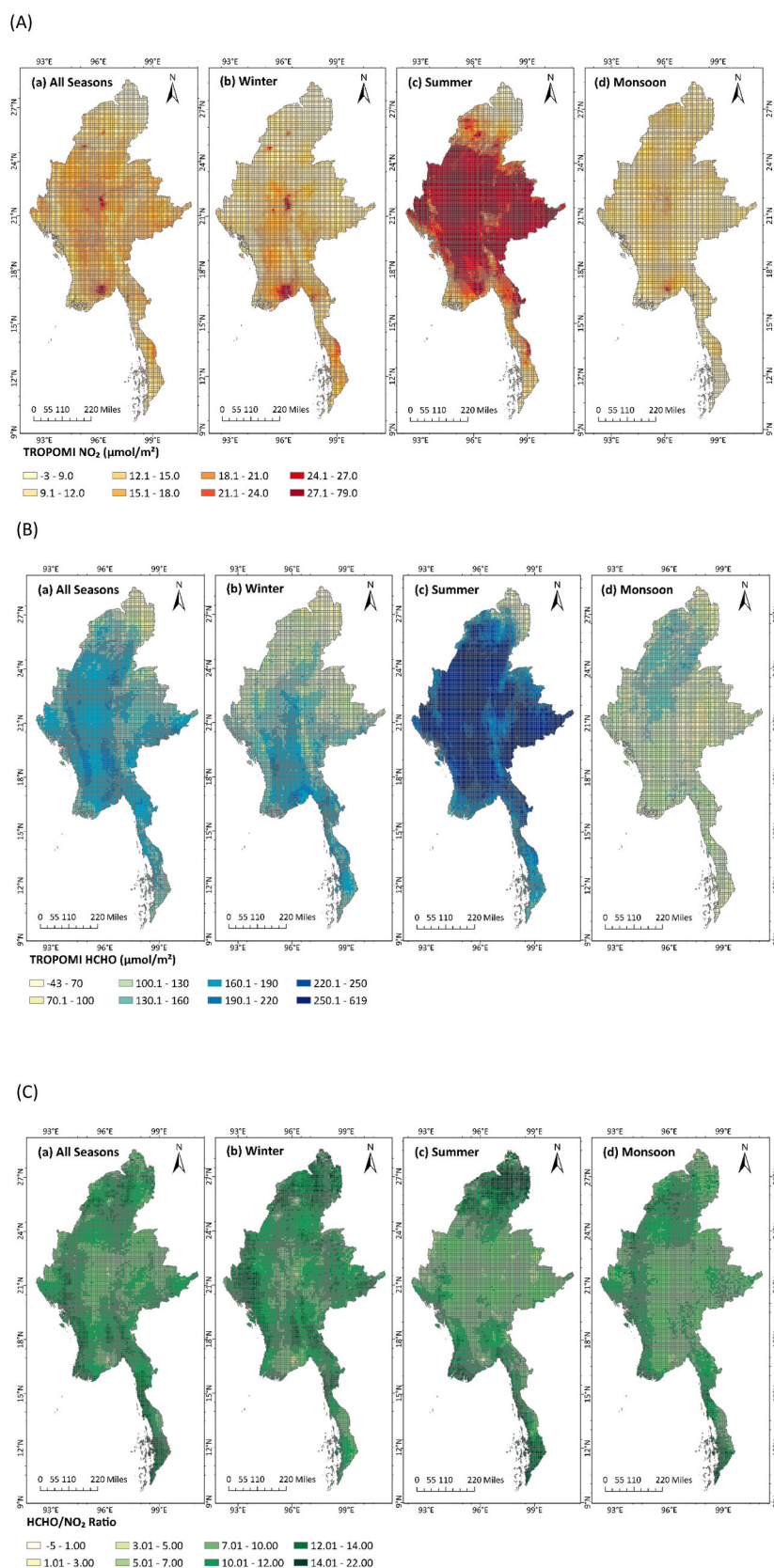


Fig. 5. Overall and seasonal spatial distributions of (A) tropospheric NO_2 , (B) tropospheric HCHO , and (C) the HCHO/NO_2 ratio over Myanmar during 2019–2024. Panels show mean values for all seasons, winter (November–February), summer (March–May), and the monsoon (June–October). NO_2 and HCHO levels are expressed in $\mu\text{mol}/\text{m}^2$, while HCHO/NO_2 ratios are dimensionless.

levels occurred in summer ($238.4 \mu\text{mol}/\text{m}^2$, $\text{SD} = 57.8$), corresponding to enhanced HCHO formation from VOCs emitted by biomass burning and biogenic sources (e.g., vegetation). In contrast, monsoon minima ($116.1 \mu\text{mol}/\text{m}^2$) were attributable to reduced VOC emissions from both biomass burning and biogenic sources, leading to decreased HCHO formation, as well as enhanced wet deposition (Goldberg et al., 2022). Spatially, summer peaks in HCHO dominated much of the country, although the northernmost regions were less affected. Local maxima in winter appeared over Lower Myanmar, likely driven by anthropogenic VOC emissions and boundary-layer trapping. The 6-year mean HCHO was highest along the Yangon–Bago–Nay Pyi Taw corridor, across the central Dry Zone, and extending northward toward Mandalay characterized by dense traffic, industrial activity, and recurrent open burning, which provide abundant VOC precursors, as well as strong sunlight that promotes secondary formation. In contrast, the southern coastal and delta regions, as well as the far northern part of Kachin State and northern Shan State, consistently exhibited lower HCHO levels, attributed to frequent rainfall and maritime inflow, and reduced anthropogenic emissions due to higher altitude (Fig. S4).

Satellite-based studies have shown that HCHO distributions in Myanmar are more diffuse than those of NO_2 , reflecting its mixed biogenic and anthropogenic sources. Biomass burning is a key driver of HCHO variability; for example, Sun et al. (2021) reported a 14.9% increase in HCHO over northeastern Myanmar in early 2020 linked to intense fire events, highlighting the sensitivity of tropospheric HCHO to episodic pyrogenic emissions (Sun et al., 2021). Dhankar and Pani (2025) also showed that seasonal HCHO peaks in Myanmar during the hot, dry months are driven by a combination of biogenic emissions, recurrent agricultural burning, and climatic factors, whereas monsoon rains reduce HCHO levels through washout, emphasizing both episodic and broader seasonal controls on regional air quality (Dhankar and Pani, 2025).

Fig. 5C illustrates the spatial patterns of TROPOMI HCHO/ NO_2 ratios, which were used as a satellite-based indicator of likely O_3 chemical regimes. The national mean ratio was 10.8 ($\text{SD} = 1.6$) across Myanmar from 2019 to 2024, with pronounced seasonality, peaking in winter (11.1) and reaching a minimum during the monsoon (10.2). From summer to winter, mean HCHO decreased by approximately 41%, whereas NO_2 declined more substantially ($\sim 47\%$). This relatively larger seasonal reduction in NO_2 contributed to the wintertime enhancement of the HCHO/ NO_2 ratio. The highest regional mean HCHO/ NO_2 ratios (approximately 12–13) were observed over the southern coastal and peninsular regions (notably Tanintharyi, Mon, and Rakhine), as well as in parts of eastern Bago and the western Ayeyarwady fringe (Ngapudaw–Labutta). Moderately elevated ratios were also found in the northern and western mountainous regions (Kachin and Chin). These regions are dominated by dense forest and mangrove ecosystems, which generate substantial biogenic VOCs and maintain low NO_2 levels due to limited anthropogenic activity, frequent precipitation, and strong marine inflow in coastal zones (Fig. S5). Conversely, the lowest regional mean HCHO/ NO_2 ratios were concentrated along the urban–industrial corridor extending from Yangon to the Mandalay Plain, where relatively high anthropogenic NO_2 emissions from dense traffic, industrial activity, and power generation suppressed HCHO/ NO_2 ratios (Fig. S3). Lower ratios were also observed in northern Shan State, likely associated with comparatively low HCHO levels.

From 2019–2021 to 2022–2024, average HCHO levels increased by 7.0% (from 153.6 to $164.4 \mu\text{mol}/\text{m}^2$), while NO_2 levels increased by 0.6% (from 14.9 to $15.0 \mu\text{mol}/\text{m}^2$). Consequently, the area-averaged HCHO/ NO_2 ratio rose by 6.7%, from 10.5 to 11.2. Declines in this ratio were confined to 13% of grid cells, primarily in urban and industrial areas experiencing greater increases in NO_2 . These consistently elevated ratios indicate that the O_3 photochemical regime across most of Myanmar was predominantly, and increasingly, NO_x -sensitive. As previously described, the HCHO/ NO_2 ratio has been widely used as a satellite-based indicator of surface O_3 sensitivity to its precursor gases.

Ratios exceeding 10 are often interpreted as indicative of strongly NO_x -sensitive conditions, particularly in rural or forested regions, whereas ratios above the commonly reported transitional range of 2.7–3.4 are generally considered indicative of NO_x -sensitive regimes, although the precise transitional range can vary regionally and seasonally (Tian et al., 2025; Finlayson-Pitts et al., 1993; Sillman, 1999). Conversely, urbanized and economically active zones such as Yangon exhibited lower mean ratios (~ 9.4), accompanied by elevated NO_2 ($20.7 \mu\text{mol}/\text{m}^2$) and HCHO ($181.8 \mu\text{mol}/\text{m}^2$), indicating a relatively weaker, yet still NO_x -sensitive regime. In addition, as discussed above, the average ratio of measured O_3 concentrations during weekends to those during weekdays in Yangon was 0.97, providing marginal evidence of NO_x -sensitive conditions. The weekend-to-weekday O_3 ratio serves as a rapid proxy for O_3 sensitivity (Sicard et al., 2020). Given the 0.1° spatial resolution of TROPOMI HCHO/ NO_2 data, transitional or potentially VOC-sensitive regimes may still occur at specific locations within Yangon. These findings suggest that NO_x emission reductions may be effective for future O_3 management across much of Myanmar, although surface precursor measurements and chemical transport modeling are needed to further evaluate local O_3 sensitivity.

3.4. Health impact assessments of $\text{PM}_{2.5}$

According to the 2019 intercensal data, Yangon's population was 7,831,830, of whom 3,843,891 (49.08%) were aged 30–90+. Within this age group, 71.1% (2,733,992) resided in urban areas, and females outnumbered males (2,148,337 vs. 1,695,554). Using the nearest ground measurements as a proxy for population exposure, the estimated population-weighted average $\text{PM}_{2.5}$ concentration for 2020–2021 was $24.6 \mu\text{g}/\text{m}^3$ (95% CI: 22.3 – $26.9 \mu\text{g}/\text{m}^3$), yielding an excess of $19.6 \mu\text{g}/\text{m}^3$ relative to the WHO guideline of $5 \mu\text{g}/\text{m}^3$. Using national age-specific non-accidental mortality data (GBD/IHME) for individuals aged 30–95+, the baseline mortality rate (Y_0) was 0.01518. The estimated average annual number of non-accidental deaths attributable to excess $\text{PM}_{2.5}$ among individuals aged ≥ 30 years in Yangon was approximately 2,106 (95% CI: 345–3,818) deaths per year. Using the alternative ChinaHEART cohort coefficient in the sensitivity analysis, the estimated avoidable mortality was 2,386 deaths per year (95% CI: 1,449–3,303 deaths per year). Although these mortality estimates were not substantially different, the estimates should be interpreted cautiously because both coefficients were derived from non-local epidemiological studies. Moreover, the 95% CIs reflect statistical uncertainty in the estimated mortality impacts, but do not fully capture uncertainty arising from methodological assumptions, including the use of non-local concentration-response coefficients, national-level mortality data, and the assumption of uniform exposure within population groups.

Due to the limited ground monitoring data and non-local assumptions, we interpreted the mortality estimates as scenario-based public health evidence derived from data-constrained conditions in Yangon and based on a WHO guideline scenario, rather than as definitive estimates of the mortality burden. Nevertheless, the public health relevance of reducing $\text{PM}_{2.5}$ exposure is supported by broader epidemiological and burden-of-disease evidence linking long-term ambient $\text{PM}_{2.5}$ exposure with premature mortality and major cardiopulmonary outcomes, including evidence from global burden assessments, WHO-related meta-analyses, and studies in low- and middle-income countries (Guo et al., 2023; Orellano et al., 2024; Institute, Health Effects Institute, 2025). Long-term $\text{PM}_{2.5}$ exposure is associated with increased hypertension prevalence and elevated systolic and diastolic blood pressure among adults and older adults, while evidence from children and adolescents indicates that $\text{PM}_{2.5}$ mass and specific components, particularly black carbon, are also associated with elevated blood pressure and hypertension risk (Song et al., 2021; Wang et al., 2025; Li et al., 2022). Short-term and personal-exposure studies further suggest that $\text{PM}_{2.5}$ exposure can affect ambulatory blood pressure and cardiovascular physiological responses, including blood pressure elevation and reduced heart rate

variability, supporting the biological plausibility of acute cardiovascular effects (Lin et al., 2021; Lee et al., 2020). Together, these findings support the broader public health benefits of reducing PM_{2.5} exposure in Yangon, while also highlighting the need for Myanmar-specific epidemiological studies, local concentration-response functions, and improved exposure assessment to refine future health burden estimates.

3.5. Limitations

This study has several limitations that should be acknowledged. First, the availability of ground-based data was limited to two monitoring sites in Yangon, as Myanmar currently lacks a nationwide ambient air quality monitoring network. Ambient O₃ measurements were available at only a single monitoring site in Yangon and met the annual completeness requirement (75%) only in 2020, thereby restricting the ability to capture interannual variability and long-term trends in O₃ pollution in the city. In addition, PM components were measured only at Station 2 and represented bulk aerosol composition rather than size-selective PM_{2.5}, thereby limiting direct comparison with PM_{2.5} mass and constraining spatial representativeness. Second, the temporal variation in air pollutants was modeled using meteorological predictors and cyclic time variables (year, month, and day of the week), without the explicit inclusion of emission inventories or traffic and activity data. This approach limits direct attribution of observed changes in pollution levels to specific emission sources or control measures and may confound meteorological effects with those driven by emissions. Therefore, the GAM results were interpreted as adjusted meteorological associations rather than isolated causal effects of meteorology alone. Third, the health impact assessment relied on multiple assumptions and external datasets. Because long-term epidemiological studies quantifying the relationship between PM_{2.5} exposure and mortality are not yet available for Yangon or, more broadly, Myanmar, the concentration-response coefficient (β) was adopted from a Bangkok-based epidemiological study of individuals aged 30–99 years. This may not fully capture health risks in younger populations or within the Myanmar context. In addition, Yangon-specific baseline mortality rates were unavailable, necessitating reliance on national-level estimates from the GBD study. Age-specific population data for Yangon were not updated for the study period; therefore, population counts were derived from LandScan raster data and adjusted using proportions from the 2019 Yangon intercensal population survey. These substitutions introduce additional uncertainty into the mortality estimates and may obscure neighborhood-level heterogeneity in exposure and risk. Lastly, this study integrated heterogeneous datasets for different analytical purposes, including ground-level air pollution measurements, meteorological reanalysis data, and satellite observations, which may introduce dataset-specific biases and uncertainties. Specifically, TROPOMI NO₂ and HCHO observations may be affected by retrieval errors and cloud screening, and the snapshot information obtained at a given satellite overpass time may not be representative of average air pollution levels on a given day, which are often represented by daily average ground-level measurements. The ERA5-Land meteorological data do not represent direct station-based observations and may not fully capture local microclimatic conditions around the monitoring sites. In addition, the limited number of ground PM_{2.5} monitors restricts the spatial representativeness of citywide PM_{2.5} exposure estimates used in the health impact assessment. Collectively, these uncertainties may be propagated differently across the analyses.

4. Conclusions

This study provides a multi-source integrated assessment of air pollution in Myanmar under data-limited conditions. We characterized air pollution dynamics, assessed satellite-informed precursor gases, and evaluated PM_{2.5}-related health impacts as evidence of public health relevance. Chronic PM_{2.5} exceedances were observed in Yangon during

2019–2024, with concentrations peaking in the dry season and decreasing during the monsoon. The GAMs showed nonlinear meteorological associations with PM_{2.5} and MDA8 O₃, reflecting atmospheric conditions related to seasonal and day-to-day variability in PM_{2.5} and MDA8 O₃ concentrations. The satellite-based HCHO/NO₂ ratios suggested that most regions of Myanmar were likely characterized by NO_x-sensitive conditions in terms of O₃ chemical regimes, which can help indicate whether NO_x or VOC emission controls would be more effective for future O₃ management. The estimated annual avoidable deaths during 2020–2021 suggest that PM_{2.5} exposure in Yangon represents a substantial public health concern. This finding highlights the need for stricter regulations and emission control technologies to reduce PM_{2.5} exposure in Yangon. Overall, the findings indicate that air pollution in Myanmar requires greater scientific and policy attention, particularly given seasonal pollution accumulation, urban-industrial emissions, biomass burning and secondary pollution influences, and limited monitoring capacity. Future work needs to strengthen ground-based monitoring capacity, develop emission inventories, apply chemical transport modeling for exposure estimation and source apportionment methods for the quantification of source contributions, and generate Myanmar-specific epidemiological evidence. By integrating ground-based, satellite, meteorological, and health-impact evidence, this study supports more informed air pollution control and public-health protection in Myanmar.

CRedit authorship contribution statement

May Myint Myat: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Hyung Joo Lee:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2026.128918>.

Data availability

Data will be made available on request.

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